AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I022*

Team Name: *4 Frames Per Second*

Domain: *Energy, Mobility and Smart Infrastructure*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Problem Statement

Noise Pollution and Reducing its Effects While Acquiring a Sustainable Form of Energy

Problem Understanding

- People in crowded spaces such as railway stations, near construction areas and industrial areas, and even more recently, AI data centers that produce a lot of high frequency noise pollution.
  The environment: lots of birds and critters

- Excess noise created from man-made means

- European Environment Agency's 2025 report says" 66,000 premature deaths, 50,000 cases of cardiovascular disease and 22,000 cases of type 2 diabetes A study by the EEA published in 2026 identifies noise pollution as a trigger for conditions such as depression and as a factor that contributes to reduced quality of life, with several other damaging effects like short term anxiety and irritability leading to long term psychological stress. Studies also show disturbed breeding and rearing cycles, psychological stress and habitat abandonment and sometimes even complete extinction of a species (eg; Chinese river dolphins, Chinese paddlefish) all resulting from noise pollution.

Existing Gaps & Challenges

- It can only be used in enclosed areas with a lot of noise (like subways, theatres, long and short tunnels, etc.) to be more efficient and work in optimal conditions.
- The cost of manufacturing, the low power output, which is not enough to support larger power consuming machinery/devices. It is not a stand-alone power source. The power is dependent on the environment's noise pollution(i.e.: the power output depends on the decibels provided by the environment).
- The existing systems face challenges for example triboelectric transducers have low resilience and collapse after only 3 million vibrations. While the other systems struggle to output more power due to low Sound Pressure Level(SPL).

Our Proposed Solution

- The core solution is a technology that has 3 main stages:
  1. Integrated hybrid resonance chamber(Quartz crystal+Helmholtz Resonance Chambers or Dual Helmholtz resonance chambers)
  2.Piezoelectric Tranducer Unit
  3.Electric Circuitry (Full wave rectifier, DC-DC connection, Battery)
- The excess noise from the environment goes into the mouth of the hybrid resonance chamber, which amplifies and tunes the sound to the resonant frequency of the piezoelectric material, and hits the piezoelectric material which causes a deformation in the material, which causes a discharge of electrons inside the material, which causes current, which is lead to a electric circuit where it is rectified and transformed and finally stored in a battery (or directly supplied to load)
- This leads to:
  1.Reduced noise pollution
  2. Better mental health, lower risk of diabetes
  3.Protected wildlife
  4.A new form of sustainable energy

Technical Approach

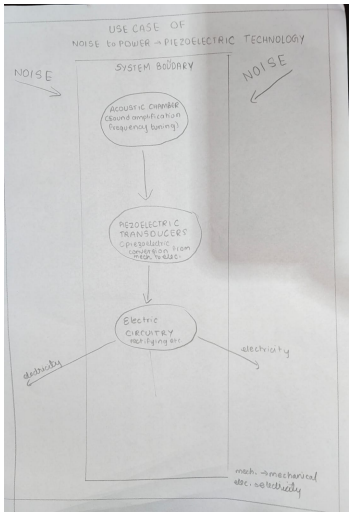


Figure: Use-case of our device

[Noise enters into a Acoustic chamber through an opening, which amplifies and tunes the sound to a specific pressure and frequency, then this pressurized noise deforms a piezoelectric material which causes current to flow, and this current goes into an electric circuit that rectifies and amplifies this current into usable energy.]

Tech Stack

Frontend: Trombone-shaped structure leading into 2 metal chambers and a battery

Backend: An acoustic amplifier, into 2 consecutive resonance chambers and an circuit board

Database: Google Scholar, Scopus, Science direct

AI/ML, APIs, or Hardware: Piezoelectric materials, circuit board, battery, Helmholtz Resonance Chambers, Quartz Crystal Resonance chambers

Innovation & Uniqueness

- A hybrid resonance chamber and an acoustic concentrator
- This approach concentrates and makes the piezoelectric material more efficient and output more power.
- 1. Higher energy output
  2.We used a hybrid resonance chamber, which was never used with piezoelectric materials

Feasibility & Scalability

- **Feasibility:** The resources needed are:
  1.Electrical circuit with a step-up transformer
  2.A hybrid resonance chamber
  3.Acoustic concentrator

- **Scalability:** The cost of production of these devices are upwards of 50,000. If this domain has more research and optimization, the device could be cheaper and more effective. The scalability is limited to the proximity to the noise.

- **Risks:** The research of this topic is very niche, hence the funding is risky.

Cost for Product (Optional)

- Development or hardware cost: The cost of production of these devices are upwards of 50,000.
- Hosting or maintenance cost: 5000 per month
- Estimated total: 75,000 (including labor charges)

Use Cases

- Use case 1: AI-Data centers- These data centers have high frequency sound wave disturb the surrounding communities and wildlife. Our devices reduce these sounds while also give back a small amount of energy back.
- Use case 2: Construction Zones- These zones are often located near highly populated areas, and carry high decibel machinery (eg. Jackhammer, Piledriver, Saw, etc.). Our solution gives these construction workers a better working conditions with reduced risk of hearing loss.
- Use case 3: Industrial Areas- The loud decibel machines that are used in industries may cause ear injuries and hearing loss. Our device here, mitigates that problem by reducing the pollution as well as giveing some energy back.

Use Cases

- Use case 4: Inside the car hood- These devices can be applied inside car hoods right next to the engine to reduce the noise pollution caused by the engines. Our device also gives back some electricity in return which can be connect to the cars battery.
- Use case 5: In Closed Dome Concerts: Closed dome concerts are loud even from the outside. Our devices can be used to reduce this noise while also storing some extra power that can be used for later.

Impacts and Benefits

- Benefits to each demographic:
  1.City dwellers: With reduced noise exposure, city dwellers will have improved mental and physical health and improved quality of life.
  2.Surrounding wildlife: they will benefit largely by their breeding and rearing cycles facing lesser interruption, leading to lesser risk of endangerment.
  3.Society has a whole will benefit;step by step, fossil fuel consumption will decrease as it gets replaced by more sustainable energy sources.

- Social impact: As exposure to loud noise decreases, people's collective chance of having stress and psychological strain will decrease, leading to better quality of life and collective mental psyche.

- Economic Impact: As piezoelectric and triboelectric substances get cheaper and our solution becomes more cost effective, more energy would be available from renewable sources (sound) and hence decreasing the economic global stress on fossil fuel availability.

- Environmental Impact: As sound pollution decreases, animals and birds would be at lesser risk of interruption and endangerment, contributing to a more sustainable and greener biosphere.

- Expected measurable outcomes would be increased public mental health and physical health, increased wildlife protection and an additional source of renewable energy.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.2	Affordable and Clean Energy	This conversion of otherwise useless noise into energy provides an alternative renewable energy source
SDG 9	Target 9.1	Industry Innovation and Infrastructure	Execution of this technology can improve human welfare in urban areas.This technology promises high in resiliency and sustainability.

Individual Member Contribution

Member	Contribution or responsibility
Dhilib S M	Researcher and Team Leader
Shyamkrishna Mahesh	Researcher and Problem Analyst
Sudeeshkumar U	Researcher and Solution analyst
Gauri Suresh	Researcher and Idea Maker

Yukti ID

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